



Universidade do Minho
Escola de Engenharia



Molecular Dynamic Simulations of Heterophilic Antibody Binding to Diagnostic IgG Antibodies

Towards Understanding Sample Interference in Rapid Immunoassays

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Point-of-Care Diagnostics

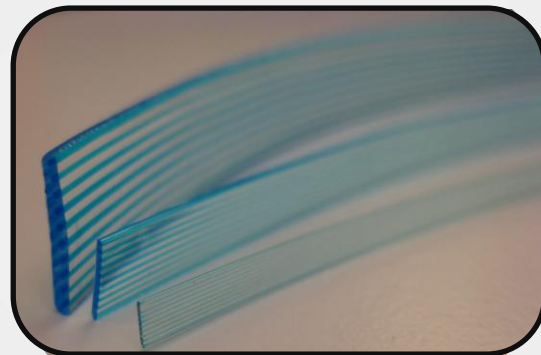
1 Contextualizing

- **Growing importance:** Fast, decentralized testing
- **The Limitation:** Current systems are qualitative

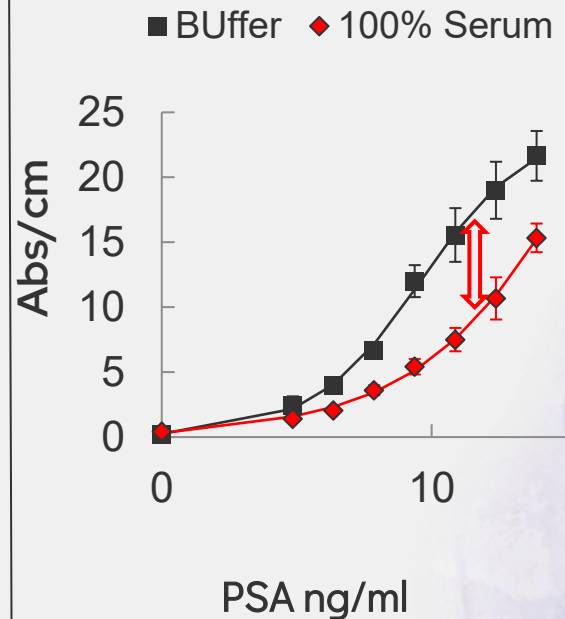
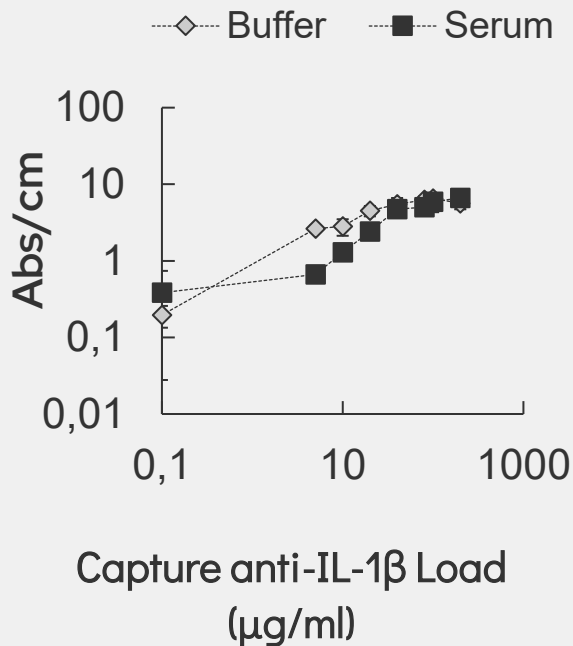
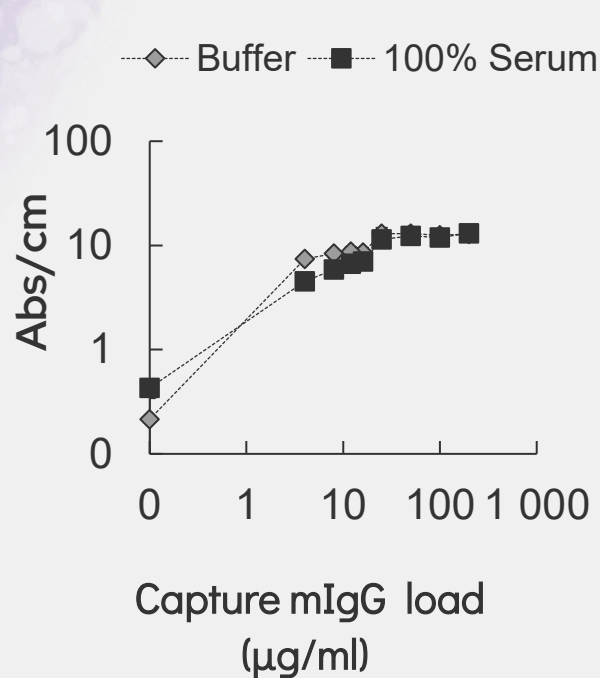


2 The Microfluidic Advantage

- **Sample Efficiency:** Requires minimal sample volumes
- **Enhanced Kinetics:** Faster antigen-antibody interactions
- **Optimization:** Reduced analysis time and lower reagent consumption



The Matrix Effect

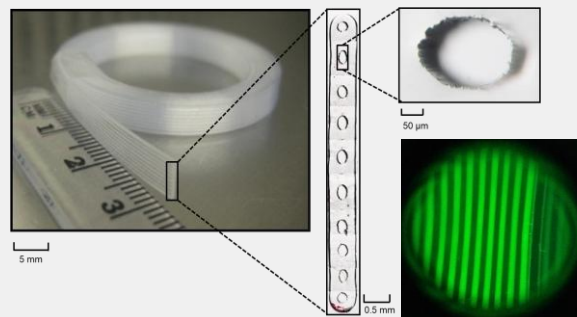
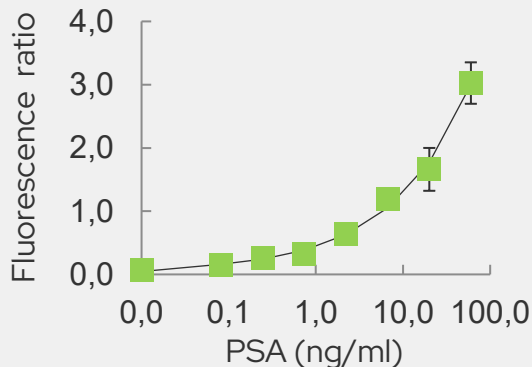


Research Questions

Can **coarse-grained molecular dynamics simulations (MD)** predict and mechanistically explain how weakly interacting serum heterophilic antibodies (HAs) disrupt target binding to surface-immobilised antibodies?

↳ Can these same MD simulations accurately describe the interaction of immobilized antibodies with the Teflon surface used in Microcapillary Films (MCF)?

↳ Antibody-Ligand Interference in Coarse-grained Environments



The microcapillary film (MCF).

Project Objectives

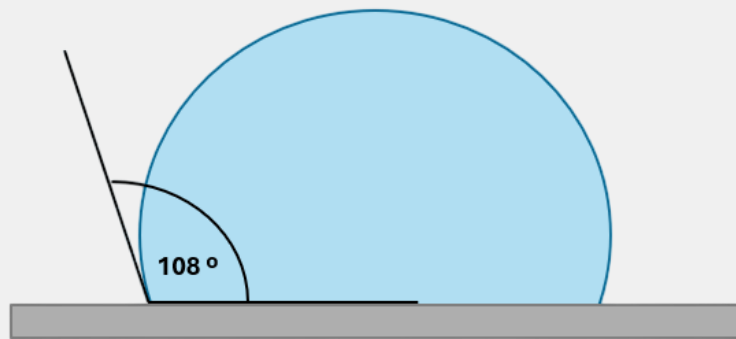
Short-term:

- Perform coarse-grained water-surface (Teflon) interactions, particularly:
 - ↳ Water-Teflon contact angle
 - ↳ Verify the stability of the system and average measures
 - ↳ Measure water-Teflon contact angle within a range of temperature
- Perform coarse-grained water-Na-Cl-surface (Teflon) interactions
- Set up a stable framework for the long-term project in Gromacs

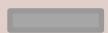
Long-term:

- Perform coarse-grained simulations of:
 - ↳ Antibody-surface (Teflon) interactions, particularly
 - ↳ Immobilised antibody and interferent antibody

Contact Angle



- Drop of Water



- Surface (Teflon)



- Contact Angle



IgG Orientation



Hydrophobicity defines protein affinity and adsorption



Fab Accessibility



Controls the efficiency of antigen binding



Wettability Measurement



Quantification of Water-Teflon interaction



Model Validation



Ensuring physical realism before introducing biological complexity

Computational Methodology

Software GROMACS

→ High-performance software for simulating molecular systems

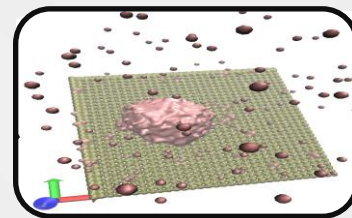
Modeling Approach: Coarse-Grained (SAFT-γ)

→ Groups of atoms are mapped into functional "beads"

→ Key Advantages: Computational Efficiency and Scalability

Outcome

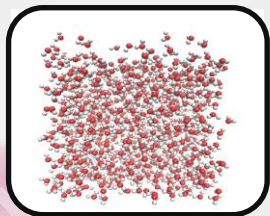
→ Provides the necessary detail to analyze biomolecule-surface interactions with optimized computational cost



System Construction

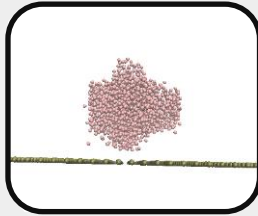
System Construction

Creation of water box and
Teflon surface via
GROMACS (coarse-
grained models (SAFT- γ))



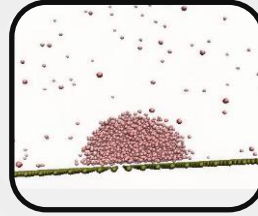
Minimization and Equilibrium

Structural relaxation
and thermal
stabilization of the
interface



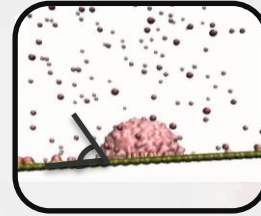
Simulation (MD)

Execution of
Molecular Dynamics
simulation



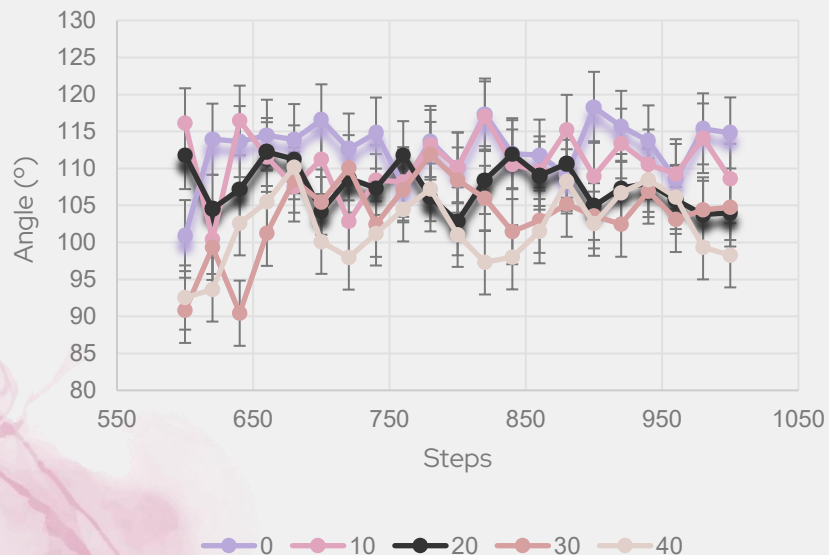
Visualization and Analysis

Frame extraction
and contact angle
calculation

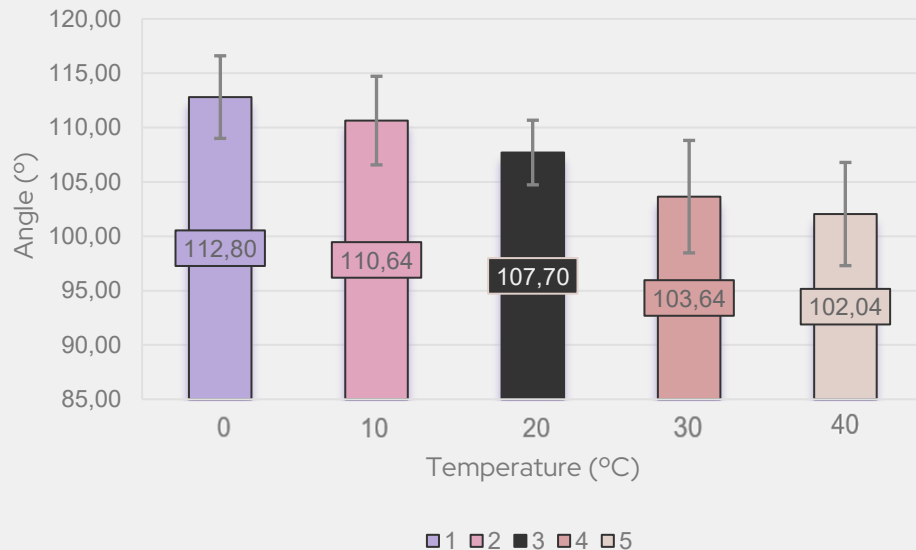


Results

Contact Angle Over Simulation Time



Average Contact Angle



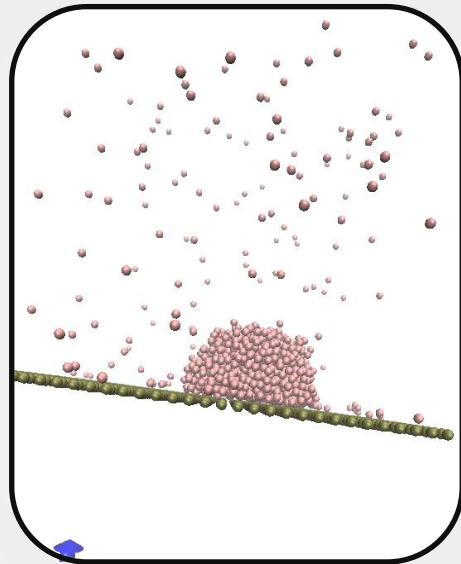
Discussion

1. Model Validation (SAFT- γ):

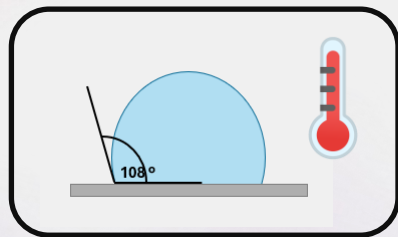
- Precise reproduction of Teflon hydrophobicity
- System ready for binding studies

2. Thermal Impact on Design:

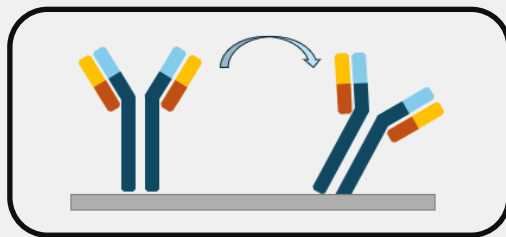
- Sensitivity of wettability to temperature
- Critical factor for biosensor efficiency and antibody orientation



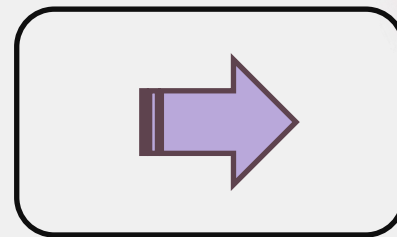
Conclusion



Model Validation



Impact of Temperature on
Wettability



Future Studies



Thank You



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Não foram usadas ferramentas
IA na elaboração desta
apresentação